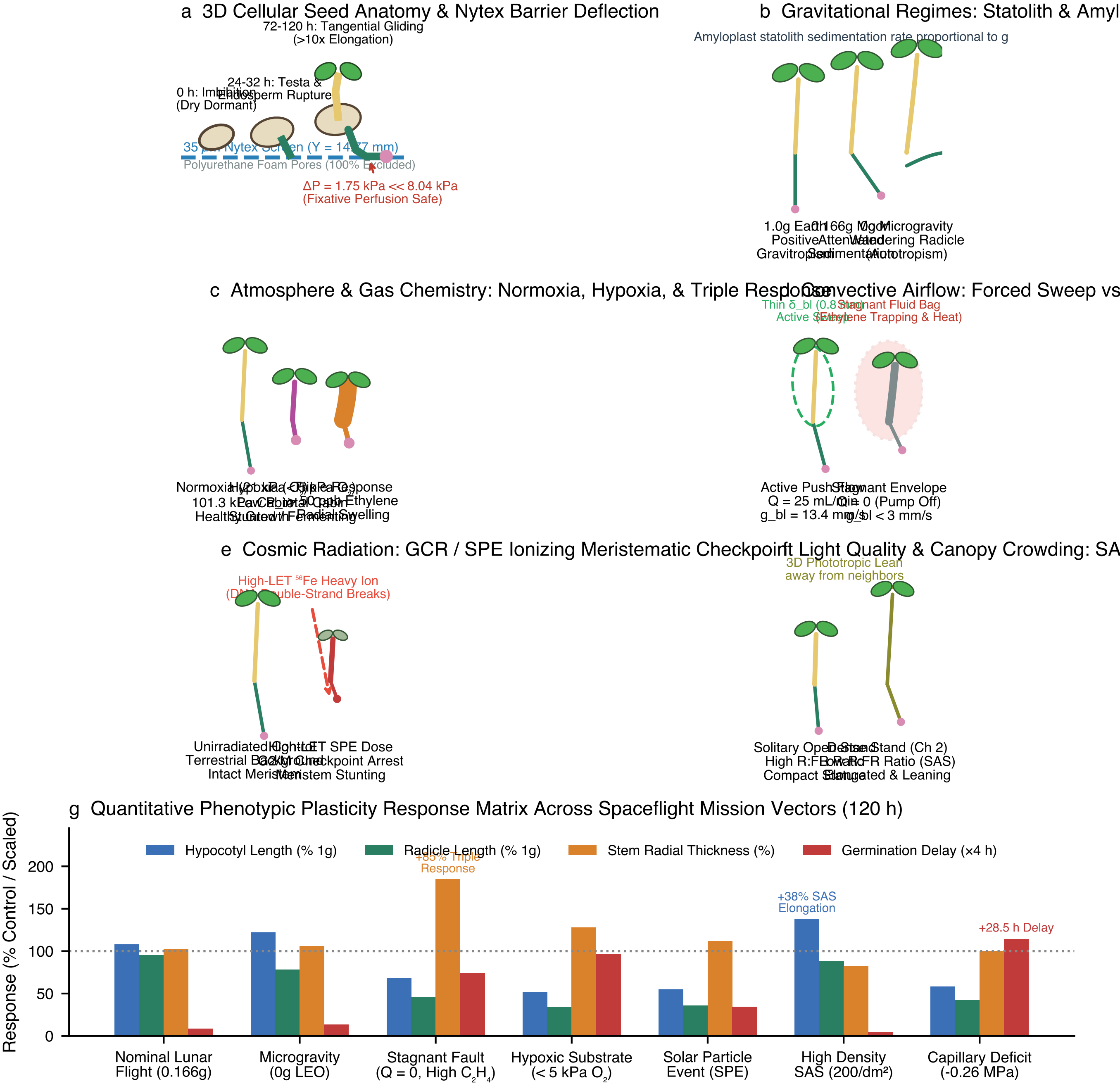
a 3D Cellular Seed Anatomy & Nytex Barrier Deflection
72-120 h: Tangential Gliding
(>10x Elongation)
24-32 h: Testa &
Endosperm Rupture
0 h: Imbibition
(Dry Dormant)
35 µm Nytex Screen (Y = 14.77 mm)
Polyurethane Foam Pores (100% Excluded)
ΔP = 1.75 kPa << 8.04 kPa
(Fixative Perfusion Safe)
b Gravitational Regimes: Statolith & Amyl
Amyloplast statolith sedimentation rate proportional to g
1.0g Earth
Positive
Gravitropism
0.166g Moon
Attenuated
Sedimentation
0g Microgravity
Wandering Radicle
(Autotropism)
c Atmosphere & Gas Chemistry: Normoxia, Hypoxia, & Triple Response
Normoxia (21 kPa O₂)
101.3 kPa Cabin
Healthy Growth
Hypoxia (< 5 kPa O₂)
Low P_total Cabin
Stunted / Fermenting
Triple Response
> 50 ppb Ethylene
Radial Swelling
Convective Airflow: Forced Sweep vs
Thin δ_bl (0.8 mm)
Active Sweep
Stagnant Fluid Bag
(Ethylene Trapping & Heat)
Active Push Flow
Q = 25 mL/min
g_bl = 13.4 mm/s
Stagnant Envelope
Q = 0 (Pump Off)
g_bl < 3 mm/s
e Cosmic Radiation: GCR / SPE Ionizing Meristematic Checkpoint
High-LET ⁵⁶Fe Heavy Ion
(DNA Double-Strand Breaks)
Unirradiated Control
Terrestrial Background
Intact Meristem
High-LET SPE Dose
G2/M Checkpoint Arrest
Meristem Stunting
f Light Quality & Canopy Crowding: SA
3D Phototropic Lean
away from neighbors
Solitary Open Stand
High R:FR Ratio
Compact Stature
Dense Stand (Ch 2)
Low R:FR Ratio (SAS)
Elongated & Leaning
g Quantitative Phenotypic Plasticity Response Matrix Across Spaceflight Mission Vectors (120 h)
Response (% Control / Scaled)
0
50
100
150
200
Hypocotyl Length (% 1g)
Radicle Length (% 1g)
Stem Radial Thickness (%)
Germination Delay (×4 h)
+85% Triple
Response
+38% SAS
Elongation
+28.5 h Delay
Nominal Lunar
Flight (0.166g)
Microgravity
(0g LEO)
Stagnant Fault
(Q = 0, High C₂H₄)
Hypoxic Substrate
(< 5 kPa O₂)
Solar Particle
Event (SPE)
High Density
SAS (200/dm²)
Capillary Deficit
(-0.26 MPa)